Elements of Quality Management

1st Seminar

Quality Planning on Project Level

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1st Seminar: Quality Planning on Project Level

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BREAK

- **Quality Plan – examples**
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What is Quality?

Criteria for buying blue jeans:

What is Quality?

Popular View:

Something vague, which can be discussed, felt and judged, but cannot be weighed or measured.

"I know it when I see it!"

Another popular view is that quality connotes luxury, class, and taste.

Professional View:

"Conformance to Requirements"

P. B. Crosby, 1979

⇒The nonconformances are regarded as defects – the absence of quality.

"Fitness for use"

Juran and Grina, 1970

⇒The products possess multiple elements of fitness for use.

Each of these elements is a **quality characteristics.**

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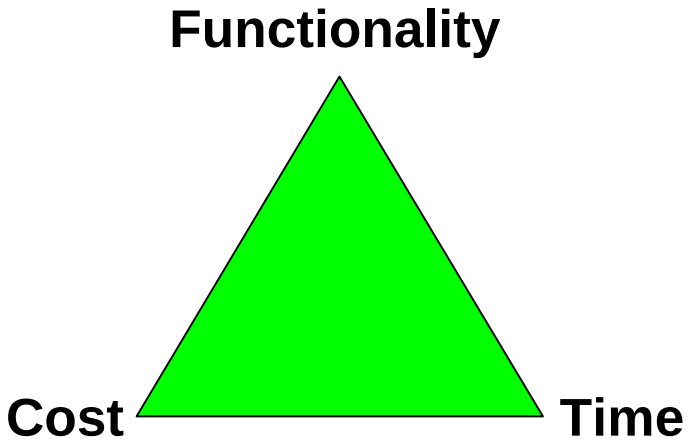
Definition of Quality

"Quality = degree to which a set of inherent **characteristics** fulfils **requirements."**

(ISO 9000:2000)



Quality means to deliver:

- the agreed **functionalities**
- at the agreed **costs** and
- in the agreed **timeframe**.

The Role of the Customer

The degree of the quality is reflected into the customer satisfaction's degree.

"Your customer are in a perfect position to tell you about quality, because that's all they're really buying. They're not buying a product. They're buying your assurances that their expectations for that product will be met.

And you haven't really got anything else to sell them but those assurances. You haven't really got anything else to sell but quality."

Guaspari, *I Know it when I see It*, 1975

Software Quality

- The goal of sw Engineering is “to produce high-quality sw”
- Question: “What is software quality?”
- Poor definition: “Something you worry about after code has been generated.”
- Common views of the sw quality:
  - Lack of bugs in the product;
  - Conformance to requirements;
  - Enhanced customer satisfaction.

=> The definition of the quality is 100% applicable to the definition of the sw quality.

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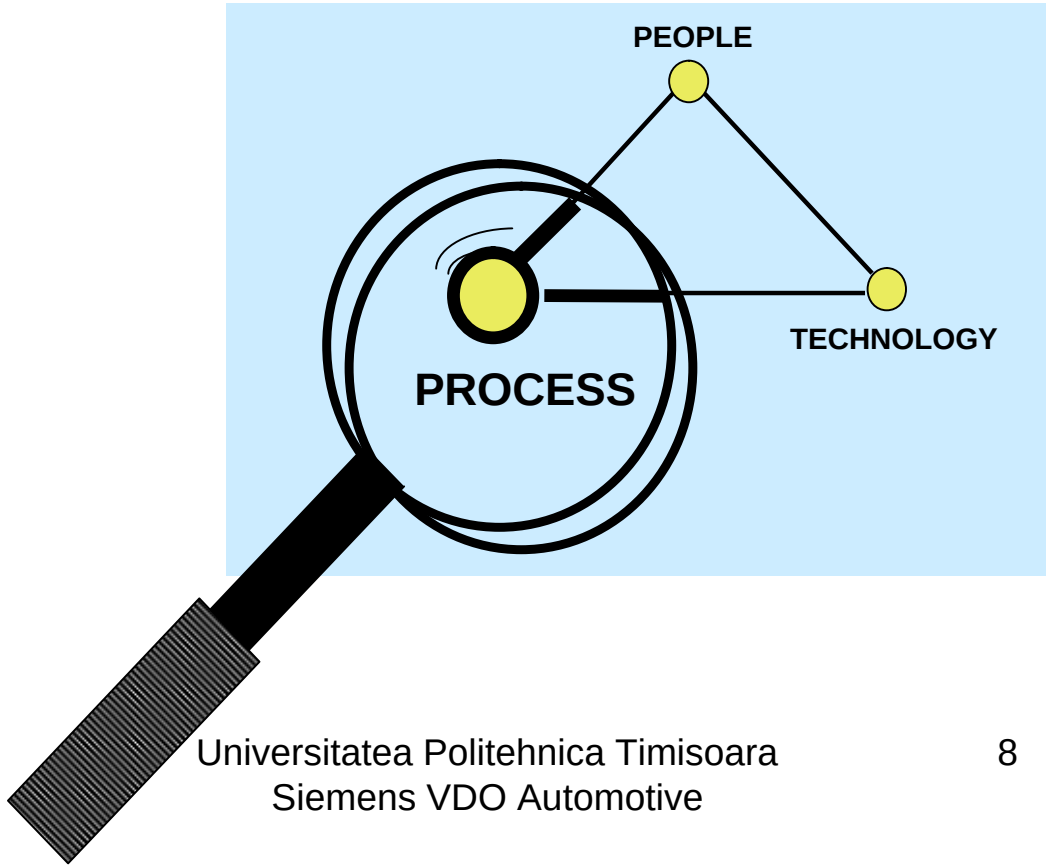
Focus on Process

The today projects are more and more complex, bigger and bigger, have strict deadlines and the market asked for smaller prices. In this context, how to assure quality work products?



The solution, as it was described by the TQM, consists in

implementing a process,

complying with it

and

continuously improving it.

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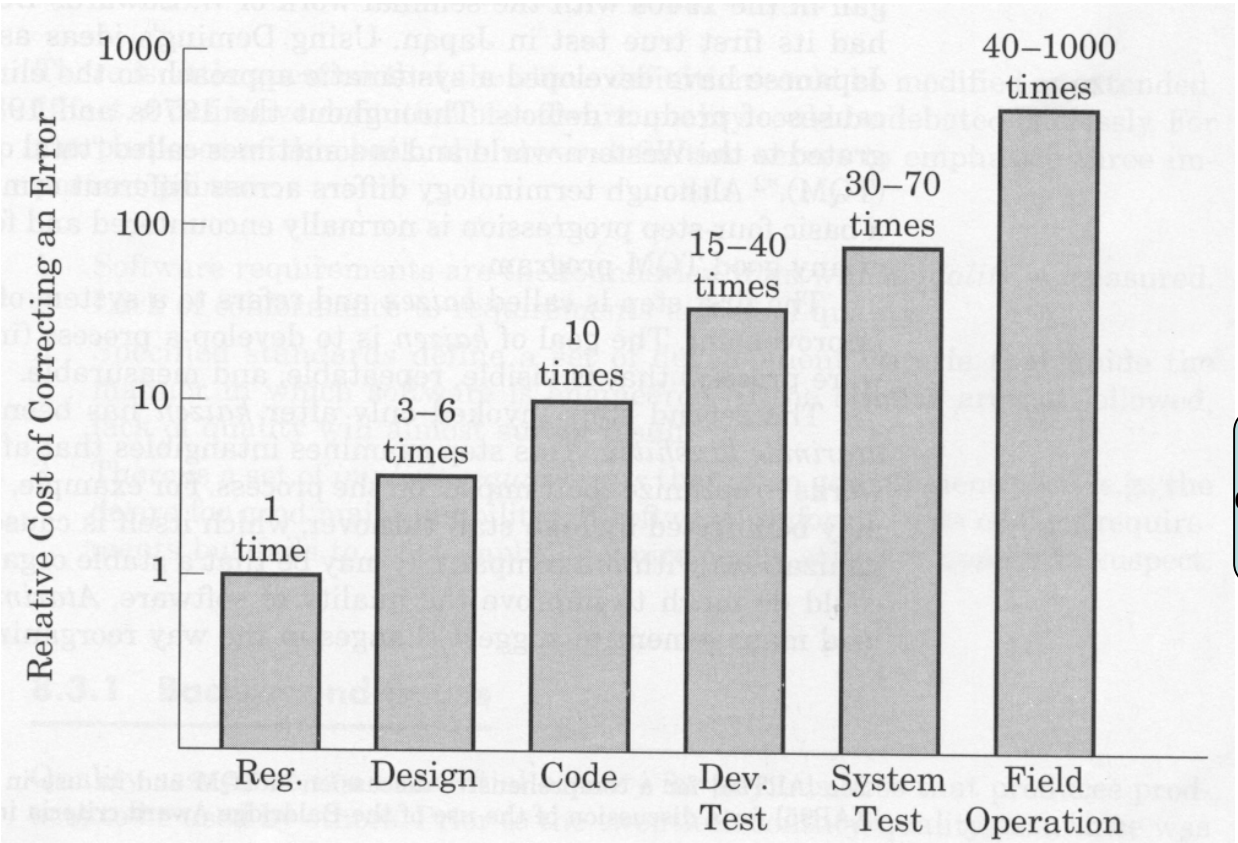
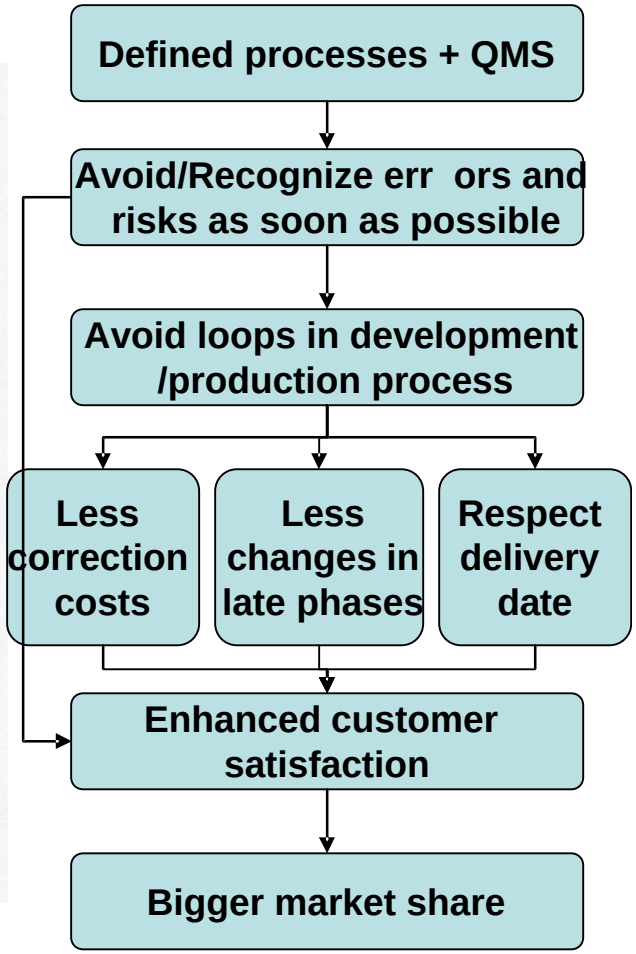
Quality Management

Do we need a QMS?





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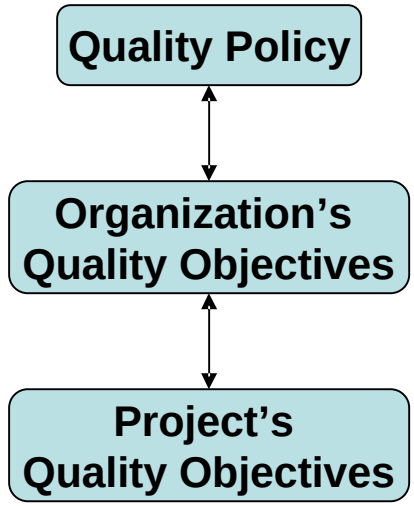
Quality Management

The purpose of Quality Management is:

- **to minimize variation**:
  - Among products, product versions;
  - Process results (defined process vs. actual process);
  - Behavior (defects noted over time);
- **to enhance customer satisfaction**.



The Quality Management System (QMS) focused on process adherence and improvement is the way to obtain the needed quality.

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Quality Management

QM includes the following processes:

- **Quality Planning**:
  - Ø Identify the quality objectives for your project and the way to attain them.
- **Quality Assurance**
  - Ø Deploy and support the implementation of the identified processes, methods and tools.
- **Quality Control**:
  - Ø Regularly monitoring the project performance to verify the status of quality objectives fulfillment and analyze the cause of the detected nonconformities and the way to remove them. It's including also the reporting.
- **Quality Improvement:**
  - Ø Increase the ability to fulfill quality **requirements**.

Roles and Responsibilities

Top Management

– Communicates to the organization the importance of meeting the requirements;
– Assures that quality objectives are defined;
– Assures the needed resources.

PL – Project Leader

– Assures the respect of the defined processes;
– Assures the quality level for the applied process and for the work products.

QE – Quality Engineer

– Plans and checks the Quality Assurance tasks;
– Checks the process adherence and evaluate work products;
– Reports – in an objective manner - the status of the quality of the process and of the products;
– Ensures that noncompliance issues are addressed;
– Supports the members of the projects in the implementation of the process and in SWQA tasks.

Members of the project teams

– Act accordingly to the defined processes;
– Commitment to the defined objectives.

Quality Management Methods and Tools

Preventative QA activities (FMEA, FTA, Lessons learned)

Reviews (object reviews and milestone reviews)

Metrics

Audits (internal audits and process audits)

Work product analysis (product audit)

Process control through direct observation

Control of non-conforming products

Quality management of suppliers

Reporting

Statistical Process Control

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The Costs of a Quality Management System

The QMS involves different costs from one organization to another, depending on the business type and the business area, but also on the chosen implementation. These costs are bigger during the implementation of the QMS and will decrease after that.

The following 3 slides show the process improvement economics as resulted from the studied of S. H. Kan on more than 10.000 sw projects. According to it sw process improvement occurs in patterns: one formal process assessment followed by a six-stage improvement program.

Stage 0: SW Process Assessment, Baseline, and Benchmark

Stage 1: Focus on Management Technologies

Stage 2: Focus on SW Processes and Methodologies

Stage 3: Focus on New Tools and Approaches

Stage 4: Focus on Infrastructure and Specialization

Stage 5: Focus on Reusability

Stage 6: Focus on Industry Leadership

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The Costs of a Quality Management System

Process Improvement Expenses $ per Capita

Stage	Meaning	Small Staff (<100)	Medium Staff (<1.000)	Large Staff (<10.000)	Giant Staff (>10.000)	Average
0	Assessment	$100	$125	$150	$250	$156
1	Management	1.500	2.500	3.500	5.000	3.125
2	Process	1.500	2.500	3.000	4.500	2.875
3	Tools	3.000	6.000	5.000	10.000	6.000
4	Infrastructure	1.000	1.500	3.000	6.500	3.000
5	Reuse	500	2.500	4.500	6.000	3.375
6	Industry Leadership	1.500	2.000	3.000	4.500	2.750
Total expenses		$9.100	$17.125	$22.150	$36.750	$21.281

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The Costs of a Quality Management System

Process Improvement Stages in Calendar Months

Stage	Meaning	Small Staff (<100)	Medium Staff (<1.000)	Large Staff (<10.000)	Giant Staff (>10.000)	Average
0	Assessment	2.00	2.00	3.00	4.00	2.75
1	Management	3.00	6.00	9.00	12.00	7.50
2	Process	4.00	6.00	9.00	15.00	8.50
3	Tools	4.00	6.00	9.00	12.00	7.75
4	Infrastructure	3.00	4.00	9.00	12.00	7.00
5	Reuse	4.00	6.00	12.00	16.00	9.50
6	Industry Leadership	6.00	8.00	9.00	12.00	8.75
Sum (worst case)		26.00	38.00	60.00	83.00	51.75
Overlap (best case)		16.90	26.60	43.20	61.42	33.64

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The Costs of a Quality Management System

Process Improvement in SW Defect Levels, Productivity, and Schedules

Stage	Meaning	Delivered Defects (%)	Development Productivity (%)	Development Schedule (%)
0	Assessment	0.00	0.00	0.00
1	Management	-10.00	10.00	-12.00
2	Process	-50.00	30.00	-17.00
3	Tools	-10.00	25.00	-12.00
4	Infrastructure	-5.00	10.00	-5.00
5	Reuse	-85.00	70.00	-50.00
6	Industry Leadership	-5.00	50.00	-5.00
Total		-95.00	365.00	-75.00

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Quality Planning

For each and every project we have to define in Quality Plan (QP):

- The quality objectives (Q targets and goals);
- Its scope (project, subproject or phases to apply to);
- The used quality processes, methods and tools and their tailoring;
- The needed resources for quality;
- The quality costs;
- The way of reporting.

The Quality Plan has to be done before to start the development and it must be updated at every modification of the project's QM.

The Quality Plan must comply the Control of Documents (must be reviewed and agreed, changes must be identified and it must be stored under CM).

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Quality Planning

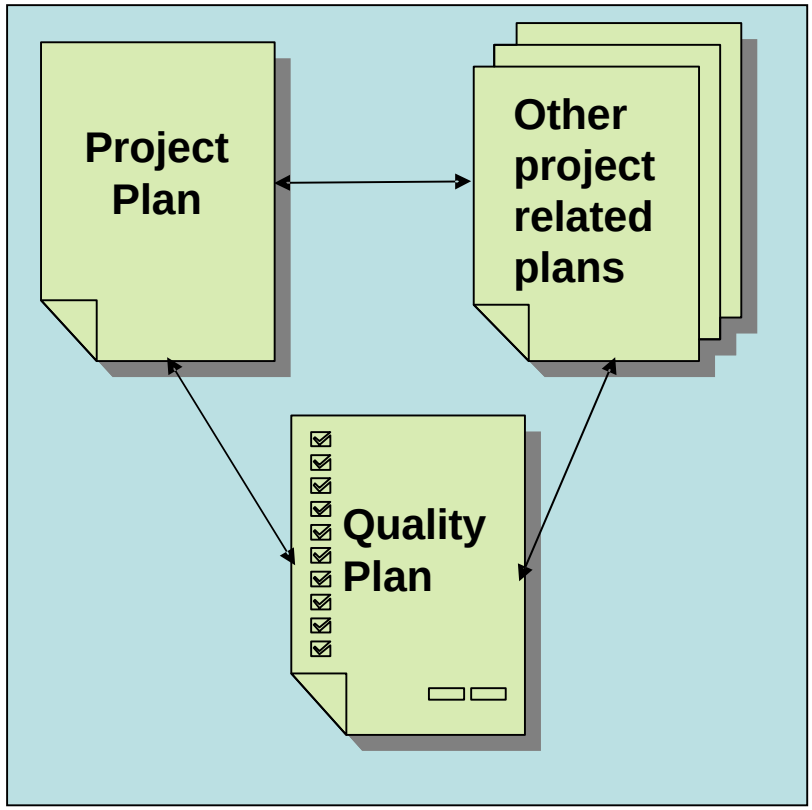
The QP has to be developed in tight conjunction with the PP and others relevant plans, like Verification and Validation Plan.

Who has to develop the QP: QE

QP has to be reviewed and signed by the PL and AL.

QP has to be communicated to the whole project team.

The tailoring of the quality processes, methods and tools applied in the organization has to be kept as minimum as possible.



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Quality Planning

Project's quality objectives:

Has to derive from the organization's quality objectives, singularized for the project. Their scope is to **enhance the customer satisfaction**.

Examples of project's quality objectives for a sw project:

Numbers of errors/10.000 ELoC (Defect rate);

Numbers of errors/function point (Defect rate);

Number of failures per n hours of operation (Reliability);

Schedule slippage (Customer satisfaction);

Actual number of reviews vs. number of planned reviews (Process adherence);

Ratio of late change request (Process adherence).

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Quality Planning

The needed resources of a QMS encompass:

- the appropriate human resources (which must have the needed skills, education and trainings);
- the needed trainings;
- the needed tools;
- the needed work environment (if something special).

All these are reflected in the quality budget, which has to be part of the project budget, clearly identifiable. But also some of them (like the needed working environment or the needed common quality trainings) will be part of the organization budget, distinct from the projects budget.

The estimation should be done per each phase of the processes which involve activities with impact on quality. This estimation should be done since beginning of the project.

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